

Mutually exclusive events



- 1 Is it possible for the following two events to happen at the same time?
 - It is a Friday.
 - It is raining.
- 2 A die is rolled. Is it possible for the following two outcomes to happen at the same time?
 - Rolling a 6.
 - Rolling an odd number.
- 3 A coin is tossed. Is it possible for the following two outcomes to happen at the same time?
 - The coin lands on heads.
 - The coin lands on tails.
- 4 Given a standard deck of cards, determine which pair of events cannot happen together:
 - Selecting a heart.
 - Selecting a 5.
 - Selecting a queen.
 - Selecting a red card.
- 5 A six-sided die is rolled. Determine which pair of events can happen together:
 - Rolling a number less than 3.
 - Rolling a 3.
 - Rolling a 5.
 - Rolling a multiple of 3.
- 6 When picking a random card from a standard pack, which two of the four events listed share no common outcomes?
 - **Event A:** picking a black card
 - **Event B:** picking a King
 - **Event C:** picking a spade
 - **Event D:** picking a club
- 7 If A and B are mutually exclusive and $P(A) = \frac{1}{4}$ and $P(B) = \frac{1}{2}$, find $P(A \cup B)$.
- 8 If A and B are mutually exclusive and $P(A) = \frac{1}{8}$ and $P(B) = \frac{7}{8}$, find $P(A \text{ or } B)$.
- 9 Two events A and B are mutually exclusive. If $P(A) = 0.37$ and $P(A \text{ or } B) = 0.73$, find



10 A random card is picked from a standard deck. Find the probability that the card is:

- a A spade or diamond.
- b A king or a 3.
- c An ace of spades or a king of hearts.

Non-mutually exclusive events

11 If $P(A) = \frac{1}{5}$, $P(B) = \frac{1}{6}$ and $P(A \cap B) = \frac{1}{30}$, find $P(A \cup B)$.

12 If $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{5}$ and $P(A \cup B) = \frac{2}{15}$, find $P(A \cap B)$.

13 If $P(A \cup B) = \frac{7}{15}$, $P(A \cap B) = \frac{1}{15}$ and $P(A) = \frac{1}{3}$, find $P(B)$.

14 If $P(A) = 0.8$, $P(B) = 0.75$ and $P(A \text{ and } B) = 0.6$, find $P(A \text{ or } B)$.

15 If $P(A) = \frac{1}{7}$, $P(B) = \frac{1}{2}$ and $P(A \text{ and } B) = \frac{1}{14}$, find $P(A \text{ or } B)$.

16 If $P(A) = \frac{1}{6}$, $P(B) = \frac{1}{4}$ and $P(A \text{ or } B) = \frac{1}{3}$, find $P(A \text{ and } B)$.

17 A random card is picked from a standard deck. Find the probability that the card is:

- a Red or a diamond.
- b An ace or a diamond.
- c An ace of spades or an ace of clubs.
- d A black or a face card.

18 A standard six-sided dice is rolled. Find the probability of rolling:

- a The number 4.
- b An odd number.
- c A 4 or an odd number.
- d An even or prime number.
- e A number that is even and prime.
- f An even number or a factor of 6.
- g A factor of 9 or an even number.
- h A number less than 3.
- i A number greater than 5.
- j a number less than 3 or greater than 5.

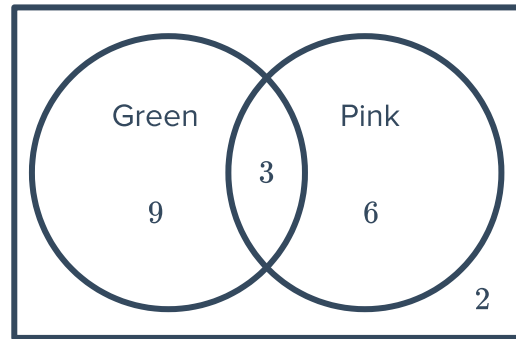
19 The Venn diagram shows the results of a

children liked:

Find the probability of a child liking:



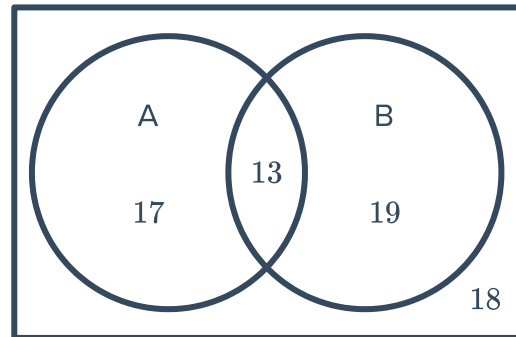
- a Green and pink.
- b Green or pink but not both.



20 Consider the Venn diagram:

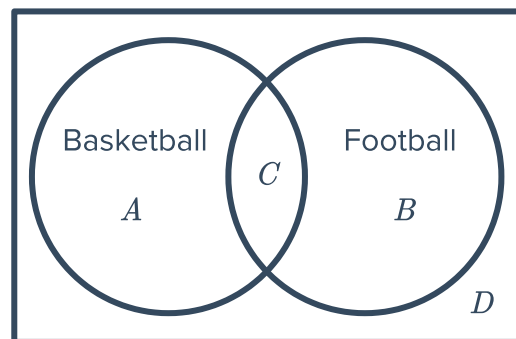
Find:

- a $P(A)$
- b $P(B)$
- c $P(\text{not } A)$
- d $P(\text{not } B)$
- e $P(B \text{ only})$



21 Out of 23 school kids, 12 play basketball and 13 play football, whilst 5 play both sports.

- a For the given Venn diagram, find the value of:
 - i A
 - ii B
 - iii C
 - iv D
- b Find the probability that a student plays football or basketball, but not both.
- c Find the probability that a student plays both football and basketball.



22 In a music school of 129 students, 83 students play the piano, 80 students play the guitar and 14 students play neither.

- a Construct a Venn diagram for this situation.
- b Find the probability that a student chosen at random plays:
 - i Both the piano and the guitar.
 - ii The piano or the guitar.

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